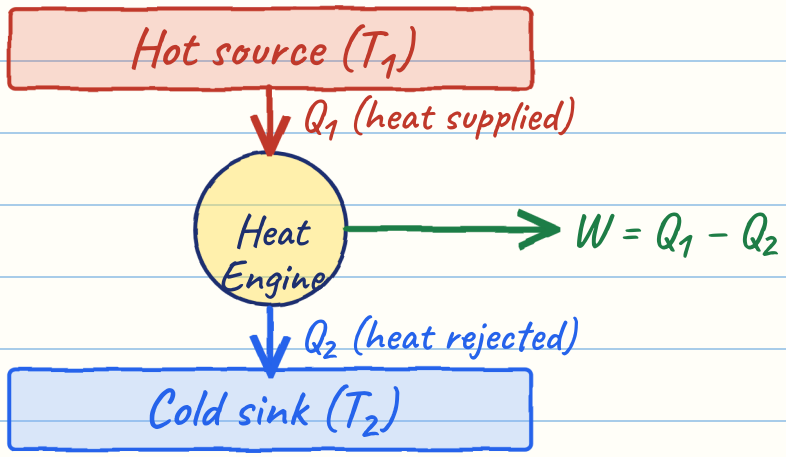
UNIT – I : I.C. ENGINES

I.1 Engine & Heat Engine

- **Engine** : a machine which converts one form of energy into mechanical work (shaft power).
- **Heat engine** : a device that converts heat energy (from burning fuel) into mechanical work continuously, using a working fluid.
- Working fluid receives heat Q_1 from a high temp. source, converts part into work W and rejects remaining heat Q_2 to a low temp. sink.

$\eta = W / Q_1$
$= 1 - Q_2/Q_1$



Thermal efficiency $\eta = W/Q_1 = 1 - Q_2/Q_1$ (always < 100%, Kelvin-Planck statement)

I.2 Classification of Heat Engines

- **External combustion (EC)** engines : fuel burns OUTSIDE the engine cylinder. Eg: steam engine, steam turbine, closed-cycle gas turbine, Stirling engine.
- **Internal combustion (IC)** engines : fuel burns INSIDE the cylinder & the products of combustion themselves act as the working fluid. Eg: petrol engine, diesel engine, gas engine, open-cycle gas turbine.

- Combustion in IC engines is intermittent (reciprocating) while in gas turbines it is continuous.

1.3 EC vs IC Engines

Point	EC engine	IC engine
Combustion	outside cylinder (boiler)	inside the cylinder
Working fluid	steam / air (separate)	products of combustion
Size & weight	bulky, heavy (boiler, condenser)	compact and light
Efficiency	low (15-25%)	high (30-40%)
Starting	slow (need steam raising)	quick starting
Fuel	any fuel, even solid	clean liquid / gas fuel
Power / weight	low	high
Cooling water	heavy (condenser)	less
Uses	power plants	automobiles, pumps, DG sets

1.4 Advantages of IC engines

- Higher overall efficiency & better power-to-weight ratio.
- Compact, light, so suitable for vehicles, aircraft, ships.
- Quick starting, low initial cost for small powers.
- Fuel is easily handled / stored; no boiler or furnace needed.

1.5 Disadvantages

- Cannot use solid fuels; needs costly refined fuels (petrol / diesel).
- Not self starting - needs a starter motor.
- High vibration & noise due to reciprocating parts.
- Exhaust gases cause air pollution (CO, HC, NOx, PM).
- Lower torque at low speeds; needs clutch & gear box.

1.6 Classification of IC Engines

(a) Based on cycle of operation

- **Otto cycle** (constant volume heat addition) - petrol / SI engines.
- **Diesel cycle** (const. pressure heat addition) - slow speed CI engines.
- **Dual cycle** (part at const. V, part at const. p) - high speed diesel engines.

(b) Based on number of strokes per cycle

- **4-stroke** engine : cycle completed in 4 strokes = 2 revolutions (720°).
- **2-stroke** engine : cycle completed in 2 strokes = 1 revolution (360°).

(c) Based on ignition

- **Spark ignition (SI)** - petrol, gas engines; spark plug ignites mixture.
- **Compression ignition (CI)** - diesel; fuel self-ignites in hot compressed air.

(d) Based on fuel used

- Petrol, diesel, gas (CNG, LPG), bi-fuel / dual fuel, alcohol (ethanol), biodiesel.

(e) Based on cylinder arrangement

- **Inline**, **V-type**, **opposed (flat)**, **radial**, **W / X / H** types.
- Vertical or horizontal; single cylinder or multi-cylinder.

(f) Based on cooling : air cooled / water cooled.

(g) Based on valve location : L, T, I (overhead), F types; OHV, OHC, DOHC.

(h) Based on speed : low (<350 rpm), medium (350-1000), high (>1000 rpm)

(i) Based on use : automobile, marine, stationary, aircraft, locomotive.

1.7 Parts of an I.C. Engine

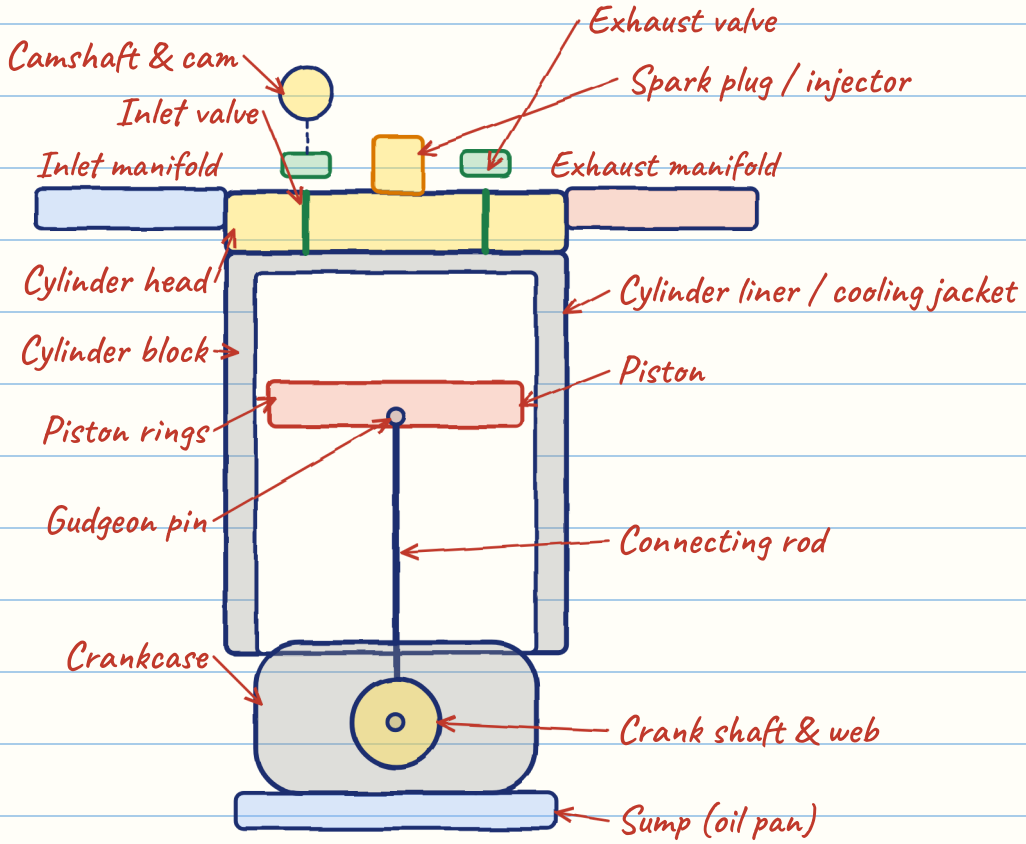


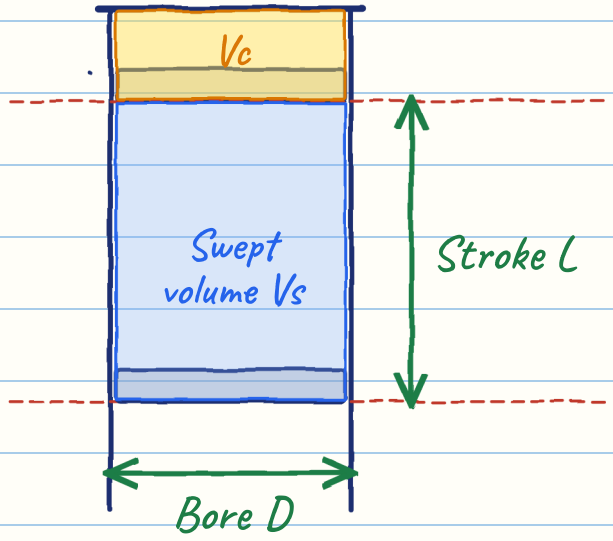
Fig: Main parts of a reciprocating IC engine (schematic)

1.7 Parts - functions (contd.)

- **Cylinder block** : main body; houses the cylinders; made of cast iron / aluminium alloy; has water jackets or cooling fins.
- **Cylinder liner** : replaceable sleeve (dry / wet) inside block, gives wear-resistant surface.

- **Cylinder head** : closes the top end; holds valves, spark plug / injector; combustion chamber formed here. Gasket seals the joint.
- **Piston** : reciprocating part; transmits gas force to the connecting rod; cast Al-alloy, light but strong.
- **Piston rings** : (a) compression rings - seal gas leakage, (b) oil control ring - scrape excess oil from wall.
- **Gudgeon (piston) pin** : joins the piston to small end of connecting rod.
- **Connecting rod** : I-section forged steel; converts reciprocating motion → rotary; big end on crank pin, small end on gudgeon pin.
- **Crankshaft** : converts to rotary motion; carries counterweights; supported by main bearings.
- **Flywheel** : heavy wheel on crankshaft; stores energy in power stroke and releases in the other strokes → smooth rotation.
- **Crankcase** : lower casing enclosing crankshaft. **Sump** holds lubricating oil.
- **Camshaft & cam** : opens inlet / exhaust valves in correct sequence; driven by timing gear / chain / belt at ½ crank speed.
- **Valves** : poppet type; inlet (large head) & exhaust (made of heat-resisting steel); spring-loaded.
- **Manifolds** : inlet manifold supplies mixture/air; exhaust manifold collects burnt gases.
- **Carburettor / Injector** : prepares the air-fuel mixture (SI) / injects fuel (CI).
- **Spark plug** : produces spark in SI engine.

1.8 Engine Terminology



Clearance volume Vc (above TDC piston)
TDC (top dead centre)

$V_s = (\pi/4)\, D^2 L$

$r = (V_s + V_c) / V_c$

$r = 1 + V_s/V_c$

BDC (bottom dead centre)
Total volume $V_1 = V_s + V_c$
Piston speed $S_p = 2LN$ (N in rps)

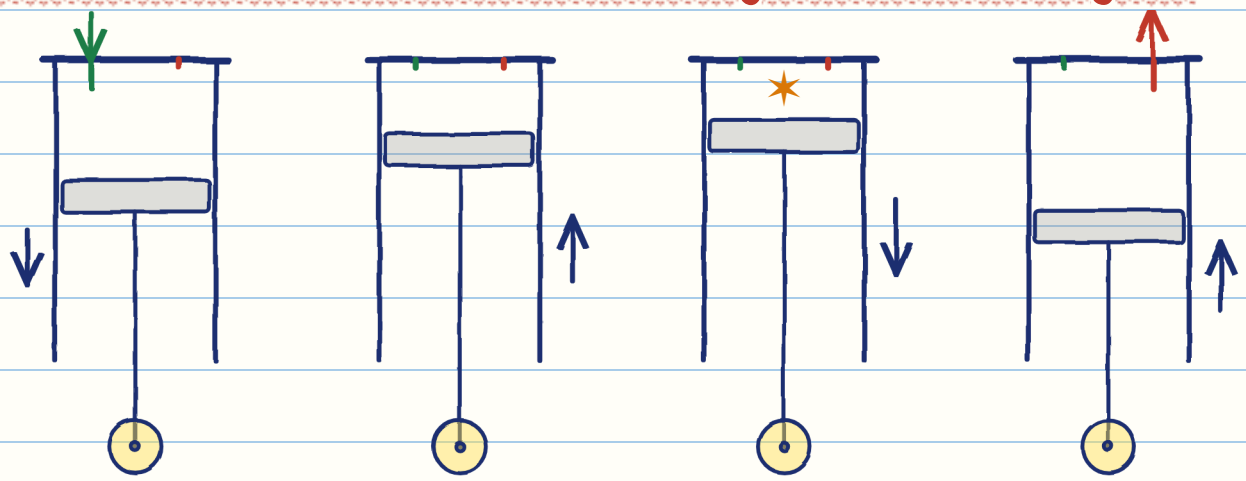
- **TDC** : extreme top position of piston (inner dead centre, IDC).
- **BDC** : extreme bottom position (outer dead centre, ODC).
- **Stroke (L)** : distance travelled by piston between TDC and BDC.
- **Bore (D)** : inside diameter of cylinder.
- **Clearance volume (Vc)** : volume above piston at TDC.
- **Swept volume (Vs)** : volume swept by piston in one stroke.
- **Compression ratio (r)** : ratio of total volume to clearance volume.
  SI : 6-10 ; CI : 15-22.

Displacement of engine = Vs × number of cylinders (cc). Other terms: Mean piston speed = 2LN.

Example

- D = 100 mm, L = 120 mm, Vc = 150 cc. Find r.
  → $V_s = (\pi/4)(10)^2(12) = 942.5$ cc ; $r = 1 + 942.5/150 = 7.28$

1.9 Four-Stroke SI (Petrol) Engine - Otto Cycle



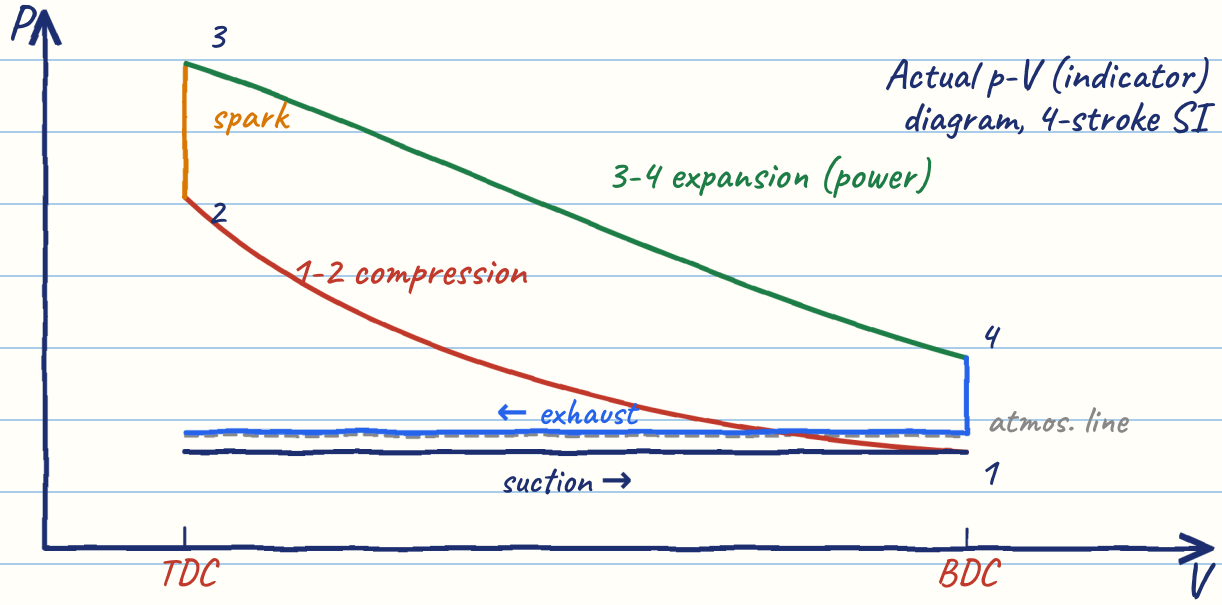
1. SUCTION 2. COMPRESSION 3. POWER 4. EXHAUST

Both valves shut in compression & power; piston: ↓ ↑ ↓ ↑ ; crank turns 720° per cycle

Working (4 strokes = 2 revolutions)

- **1. Suction stroke** : inlet valve open, exhaust closed; piston goes TDC → BDC; air-fuel mixture (from carburettor) drawn in at about 0.8-0.9 atm.
- **2. Compression stroke** : both valves closed; piston BDC → TDC; mixture compressed to ≈ 8-10 bar; near the end (≈ 20-30° before TDC) the spark plug fires.
- **3. Power / expansion stroke** : combustion raises pressure to 35-45 bar, T ≈ 2500 K; gases push piston TDC → BDC; **only stroke producing work**. Exhaust valve opens near the end (EVO).
- **4. Exhaust stroke** : exhaust valve open; piston BDC → TDC pushes burnt gases out at ≈ 1.1-1.2 atm.

1.9 (contd.) p-V diagram of 4-stroke SI engine



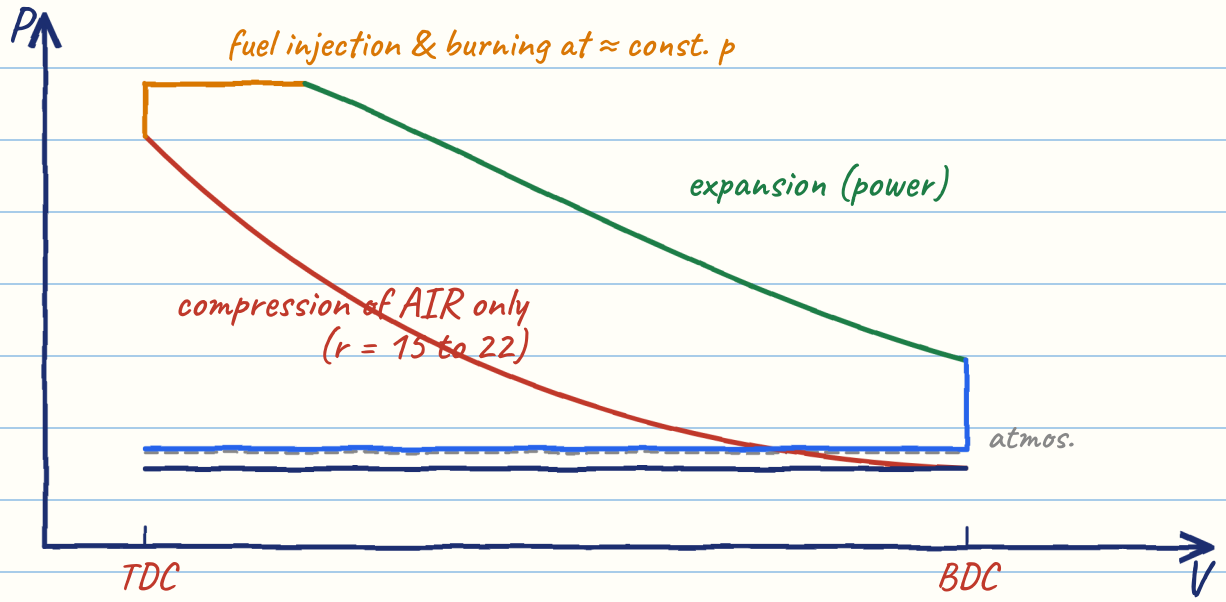
- Area of upper loop = **work done** (positive); lower loop (suction & exhaust) = pumping work (negative), small.
- Net work = Area(upper) − Area(lower).
- **Heat is added at nearly CONSTANT VOLUME** (2→3) because combustion is rapid.
- Only 1 power stroke in every 2 revolutions ⇒ power strokes per minute = N/2.

Valve operation in 4-stroke engines

- Each valve opens once in 2 revs, so camshaft turns at **half** the crank speed.
- Cam, follower, push rod, rocker arm, valve spring - valve train.
- Gear ratio crank : cam = 2 : 1.

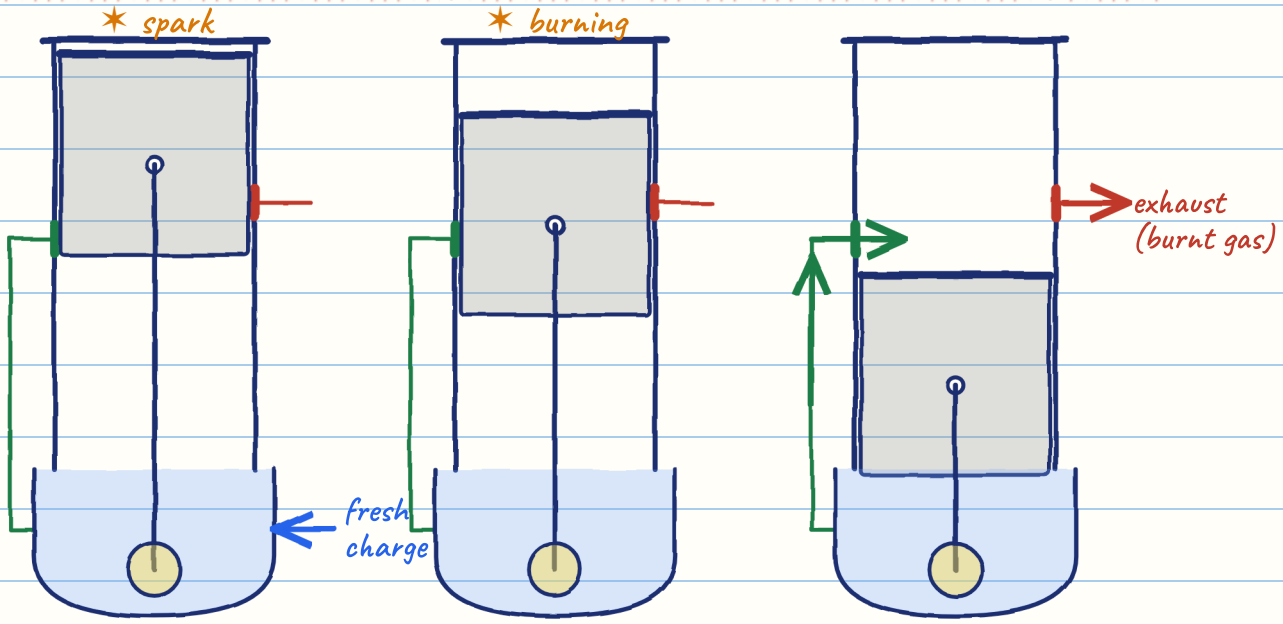
1.10 Four-Stroke CI (Diesel) Engine

- Intake of **air only** (no throttle) → compressed to 30-40 bar, T = 600-800 °C; fuel injected near end of compression; self-ignites ⇒ no spark plug.



- **Suction** : only air enters (inlet open). **Compression** : high compression ratio 15-22 raises temperature above self-ignition temp. of diesel.
- **Power** : fuel (≈ 10-15° bTDC) injected at 150-250 bar; burns at roughly const. pressure; expansion pushes piston.
- **Exhaust** : burnt gases pushed out.
- Load is controlled by *quantity of fuel* (quantity governing), not by throttling air.

1.11 Two-Stroke Engine (crankcase scavenged)



(a) Upstroke: compression + crankcase suction

(b) Downstroke: power + crankcase compression

(c) Exhaust & transfer (scavenging)

- Cycle in **one revolution** : power stroke at every revolution → 2 strokes (up & down). No valves - ports (transfer, exhaust, and inlet - piston-port or reed valve) are covered / uncovered by the piston.
- **Upward stroke (a)** : piston covers transfer & exhaust ports → compresses the charge above; at the same time creates vacuum in crankcase → fresh charge enters through inlet port. Spark near TDC.
- **Downward stroke (b)** : combustion → power stroke; piston covers inlet port & compresses charge in crankcase (≈ 1.3-1.5 bar).
- (c) Exhaust port opens first (blow-down), then transfer port opens → pressurised charge from crankcase flows to cylinder, deflector on piston pushes burnt gas out (**scavenging**).

1.12 Scavenging in 2-stroke engines

- Scavenging = process of sweeping out burnt gases with fresh charge.
- Types : (1) **Cross-flow** - deflector piston; (2) **Loop** - ports on same side, charge loops (Schnuerle / Curtis); (3) **Uniflow** - charge enters at one end & exhaust at other end through valves/ports (best, used in large diesels).
- Problem in petrol 2-stroke : part of fresh charge escapes into the exhaust ⇒ fuel loss, HC emissions, so fuel injection is used now.

Comparison : 4-stroke vs 2-stroke

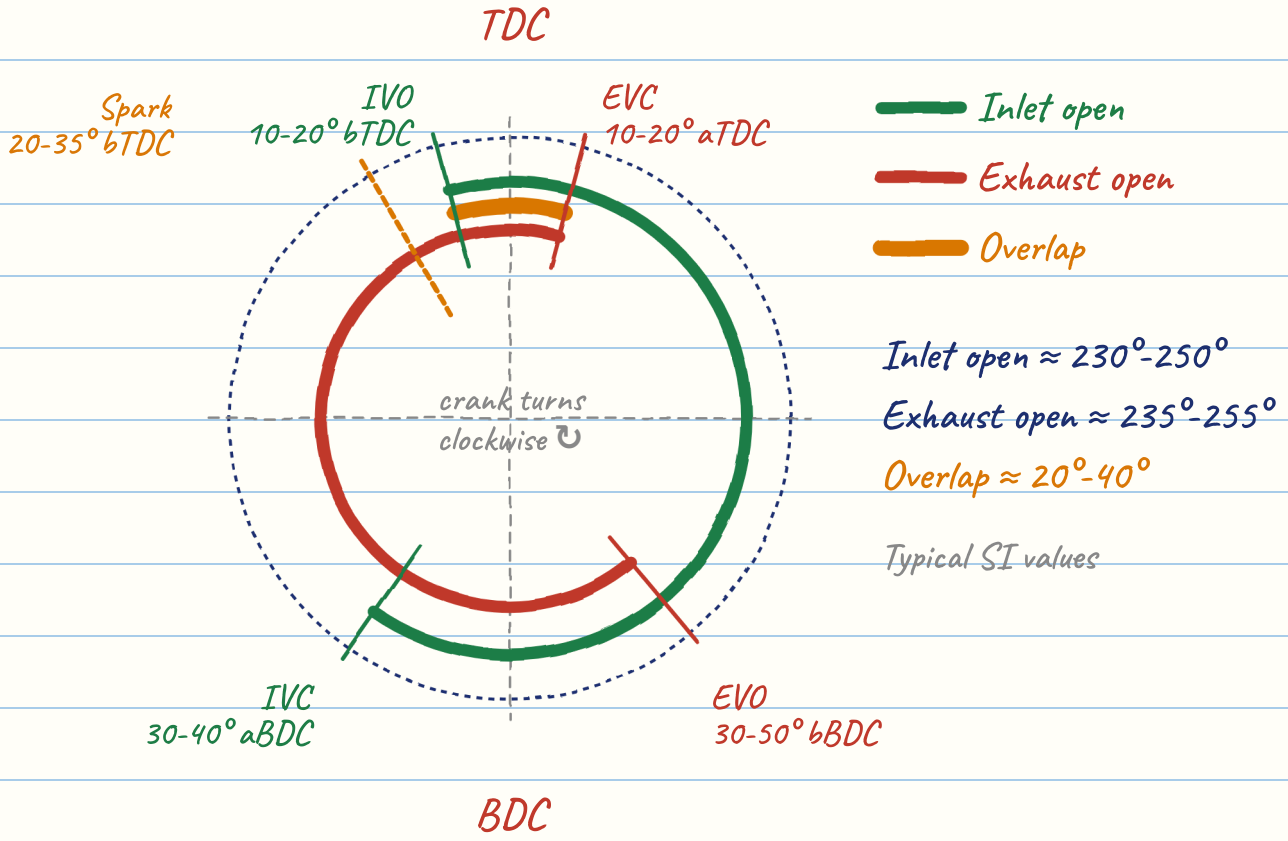
Point	4-stroke	2-stroke
Cycle	4 strokes / 2 revs	2 strokes / 1 rev
Power stroke	once in 2 revs	once every rev
Power for same size	less	≈ 1.6-1.8 × more (not 2×)
Valves	poppet valves + cam	ports only (simple)
Flywheel	heavy	light
Thermal efficiency	higher	lower (fuel loss)
Fuel consumption	lower	higher
Lubrication	sump / pressure	petrol-oil mix
Cooling	easier	harder (more heat)
Wear	less	more
Used in	cars, trucks, buses	scooters, mopeds, outboards

1.13 SI Engine vs CI Engine

Point	SI engine	CI engine
Cycle	Otto (const. volume)	Diesel / Dual
Fuel	petrol (volatile, high self-ignition temp.)	diesel (less volatile, low SIT)

Point	SI engine	CI engine
Intake	air + fuel mixture	air only
Fuel supply	carburettor / MPFI	fuel injection pump & injector
Ignition	spark plug (electrical)	self ignition by hot air
Compression ratio	6 - 10	15 - 22
Max. pressure	35 - 45 bar	60 - 100 bar
Weight	lighter	heavier, stronger
Efficiency	25 - 30 %	35 - 40 %
Speed	high (up to 6000+ rpm)	lower (up to 4000)
Load control	quantity of mixture (throttle)	quantity of fuel
Running cost	higher	lower
Noise / vibration	less	more
Emission	CO, HC	NOx, smoke, PM
Starting	easy	harder in cold

1.14 Valve Timing Diagram (4-stroke)



- It shows the **angular position of crank** (w.r.t. TDC & BDC) at which valves open and close.

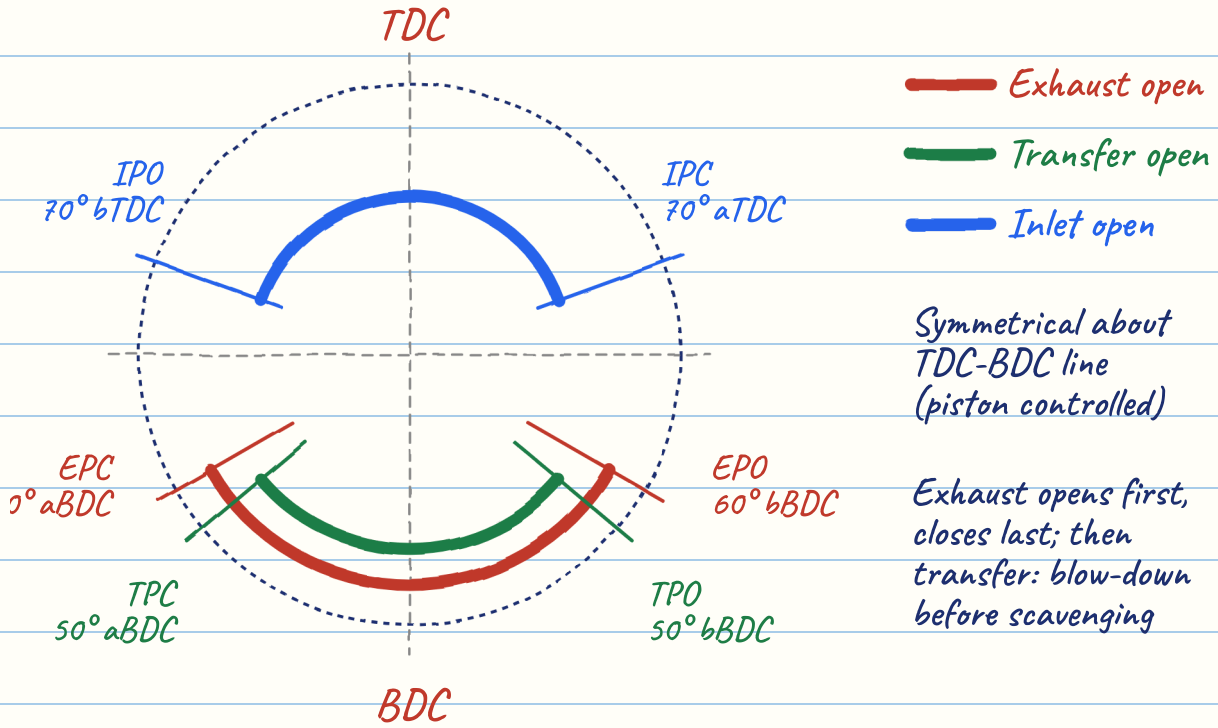
1.14 Valve timing (contd.)

- **Actually**, inlet valve opens **before TDC** (IVO, 10-20°) - **lead** - so that it is fully open when the suction stroke starts, giving full charge.
- Inlet valve closes **after BDC** (IVC, 30-40°) - **lag** - since the incoming charge still has momentum (ram effect), more charge enters even though the piston is rising.
- Exhaust valve opens **before BDC** (EVO, 30-50°) - **lead** - so that pressure falls (blow-down) and the piston has less back pressure in exhaust stroke.
- Exhaust valve closes **after TDC** (EVC, 10-20°) - **lag** - to get complete scavenging by inertia of exhaust gas.

Valve overlap = (IVO lead + EVC lag) ≈ 20-40° when both valves are open near TDC : good scavenging, better volumetric efficiency; but in petrol engine too large overlap wastes fuel.

- **Effect of speed** : at higher speeds greater lead and lag are given because time available is less.
- Diesel values : IVO 10-25° bTDC, IVC 25-40° aBDC, EVO 35-45° bBDC, EVC 10-20° aTDC, injection 10-15° bTDC.
- Diesel engine overlap is larger (pure air escapes, no fuel loss); also helps cooling of valves/piston crown.
- Ignition (SI) 20-35° before TDC and fuel injection (CI) 10-15° before TDC are also marked on the diagram.

I.15 Port Timing Diagram (2-stroke)



- Shows the crank angles at which the **ports** open & close for piston-controlled 2-stroke engine.
- **Exhaust port** opens earlier (EPO ≈ 60° bBDC) and closes later (EPC ≈ 60° aBDC) than the transfer port (TPO/TPC ≈ 50°).

- *Inlet port* opens ≈ 70° bTDC and closes ≈ 70° aTDC (as piston covers it).
- Diagram is symmetrical because the piston itself uncovers & covers ports (fixed timing) - for unsymmetrical timing, reed / rotary valves are used.

Quick Revision

- $r = 1 + Vs/Vc$; $Vs = (\pi/4)D^2L$; 4-S: 720° ; 2-S: 360°
- SI → Otto, spark, r = 6-10 ; CI → Diesel, auto-ignition, r = 15-22